Practical 20: Threads and Handlers

Source: MODULE 2.pdf | Source pages: 80–83

Aim

To develop and test an Android application demonstrating threads and handlers.

Steps to Perform the Practical

1. Create an Android Studio project for the threading practical.

2. Add the UI components shown in the source layout.

3. Create a background Thread for the time-consuming work.

4. Use a Handler to send results or updates back to the main UI thread.

5. Run the app and trigger the background task.

6. Verify that the UI remains responsive and that the result is displayed.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

Concurrency, Media and Security

13. Develop an Android application demonstrating Threads and Handlers for background processing.

Project Name: ThreadHandlerDemo

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
       android:id="@+id/btnStart"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Start Background Task"/>
   <TextView
       android:id="@+id/txtResult"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:textSize="22sp"
       android:paddingTop="20dp"/></LinearLayout>
```

MainActivity.java

```
package com.example.threadhandlerdemo;

import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnStart;
    TextView txtResult;

    Handler handler = new Handler(Looper.getMainLooper());

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnStart = findViewById(R.id.btnStart);
        txtResult = findViewById(R.id.txtResult);
```

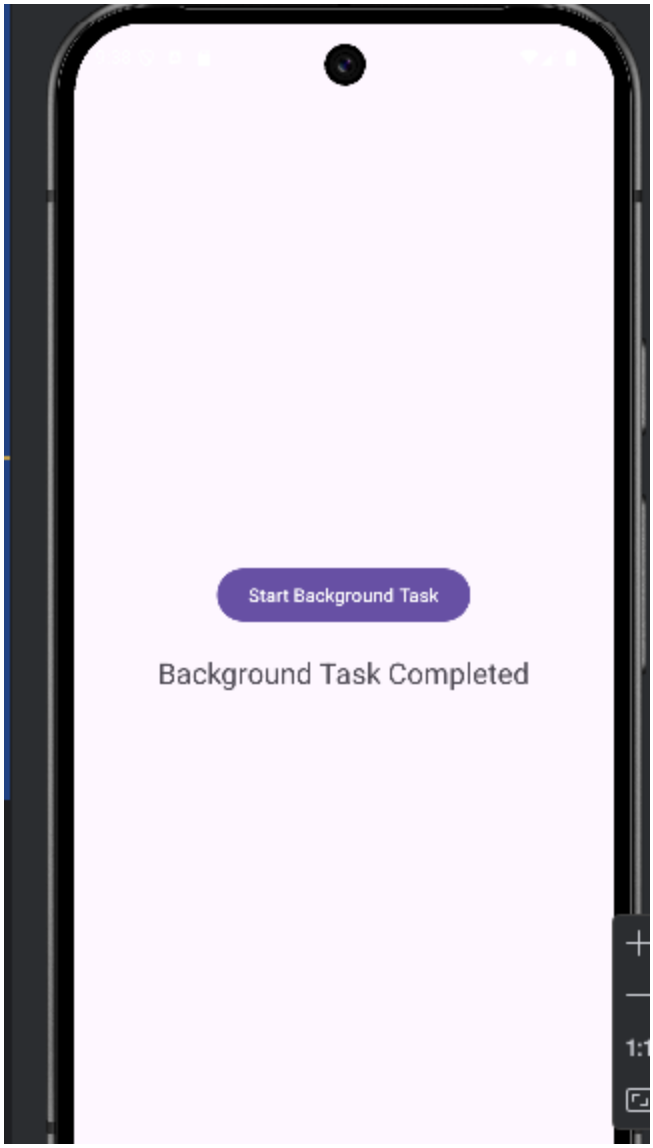
```
        btnStart.setOnClickListener(v -> {


            // Background Thread
            Thread thread = new Thread(() -> {
                try {
                    // Simulate Background Work
                    Thread.sleep(3000);


                } catch (Exception e) {
                    e.printStackTrace();
                }


                // Update UI using Handler
                handler.post(() -> {
                    txtResult.setText("Background Task Completed");
                });
            });


            thread.start();
        });
    }
}
```


Start Background Task
Background Task Completed